

Introduction

We are provided a brain

System identification

The exact Hankel singular-value spectrum, obtained via same-seed impulse subtraction (a theoretical reference not physically realisable with a real brain), has rank 6 (Figure 1). A physically realisable single-run OLS estimate ($N = 5,000$ steps) raises a noise floor that obscures modes 3–6 (exact values $\sigma_3/\sigma_1 = 0.125$ down to $\sigma_6/\sigma_1 = 3.3 \times 10^{-3}$); BIC model selection recovers only $n_x = 2$. These modes carry little input–output energy and are near-unobservable from any finite honest recording. However, the downstream decoder operates on frequency-selective latent activity, so a model that misrepresents the transfer function will produce incorrect muscle activations regardless of one-step prediction quality. ERA+EM was therefore fitted at $n_x \in \{2, 4, 6\}$ to identify the minimum sufficient order.

At $n_x = 2$ the model reproduces per-neuron frequency selectivity for u_1 but not for u_2 (Figure 5). At $n_x = 4$ both input channels are recovered; $n_x = 6$ yields no visible further improvement. ERA+EM at $n_x = 4$ was therefore selected. The Bode magnitude error (Figure 3), computed diagnostically via same-seed impulse subtraction, confirms this pattern: mean error falls substantially from $n_x = 2$ to $n_x = 4$, with negligible reduction at $n_x = 6$.

CVA+EM was fitted at $n_x = 4$ for comparison. Its transfer function reproduces the u_1 selectivity but misrepresents u_2 relative to the honest OLS estimate (rightmost column, Figure 5). One-step-ahead Kalman prediction on $N = 100$ held-out trials ($T = 2,000$ steps each) confirmed ERA+EM superiority across all output channels (Figure 4). ERA+EM at $n_x = 4$ was adopted; the identified matrices are shown in Figure 2.

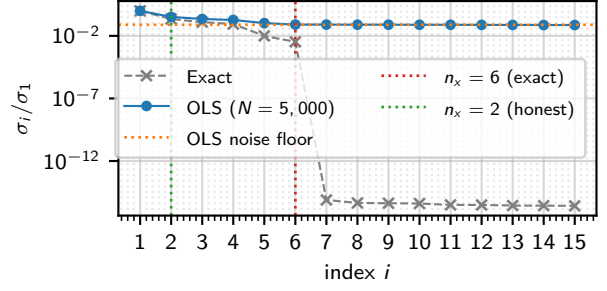


Figure 1: Normalised Hankel singular values σ_i/σ_1 for the block Hankel matrix (40 block rows, 60 block columns, $n_u = 2$, $n_y = 16$). Grey dashed: exact values via same-seed impulse subtraction (not physically realisable; theoretical reference only). Blue solid: OLS estimates from a single honest 5000-step recording. Orange dotted: OLS noise floor (median of OLS values above exact rank). Red dotted: $n_x = 6$ (exact). Green dotted: $n_x = 2$ (honest, BIC model selection). Modes 3–6 lie below the honest noise floor, rendering them undetectable from finite recordings. [Source](#)

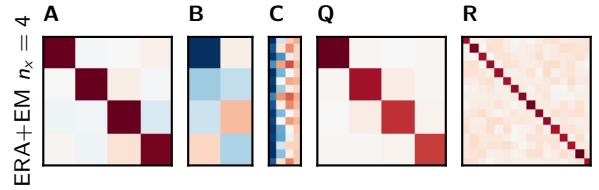


Figure 2: Identified system matrices for ERA+EM ($n_x = 4$). Red indicates positive, blue negative; colour scales are symmetric about zero. [Source](#)

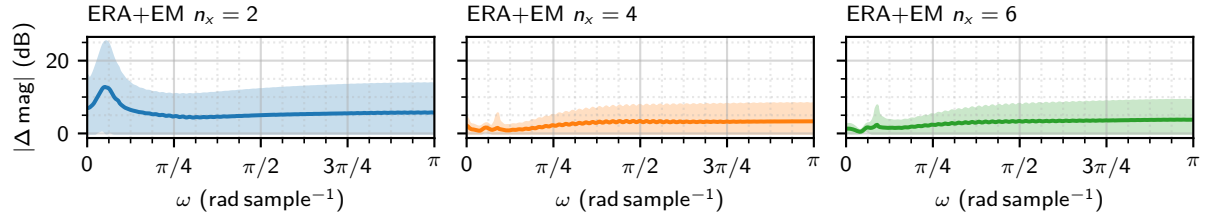


Figure 3: Magnitude error of ERA+EM transfer functions for $n_x \in \{2, 4, 6\}$ (left to right). Each panel shows the absolute magnitude error (dB) between the exact empirical transfer function and the analytic model response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$, as mean ± 1 s.d. across all $n_u n_y = 32$ output–input channels. The empirical reference is the exact discrete-time frequency response obtained via impulse subtraction (same-seed noise cancellation), DFT zero-padded to 512 points. Panels share a common y-axis. [Source](#)

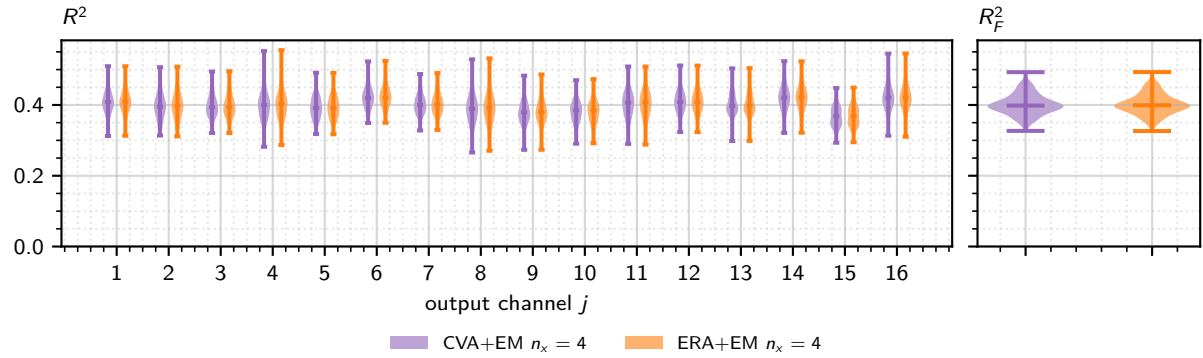


Figure 4: Per-channel R^2 (left, $n_y = 16$) and Frobenius R_F^2 (right) of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at $n_x = 4$, across $N = 100$ held-out trials of $T = 2000$ steps (first 100 steps excluded as Kalman transient). Median: CVA+EM $n_x = 4$ $R_F^2 = 0.398$; ERA+EM $n_x = 4$ $R_F^2 = 0.399$. [Source](#)

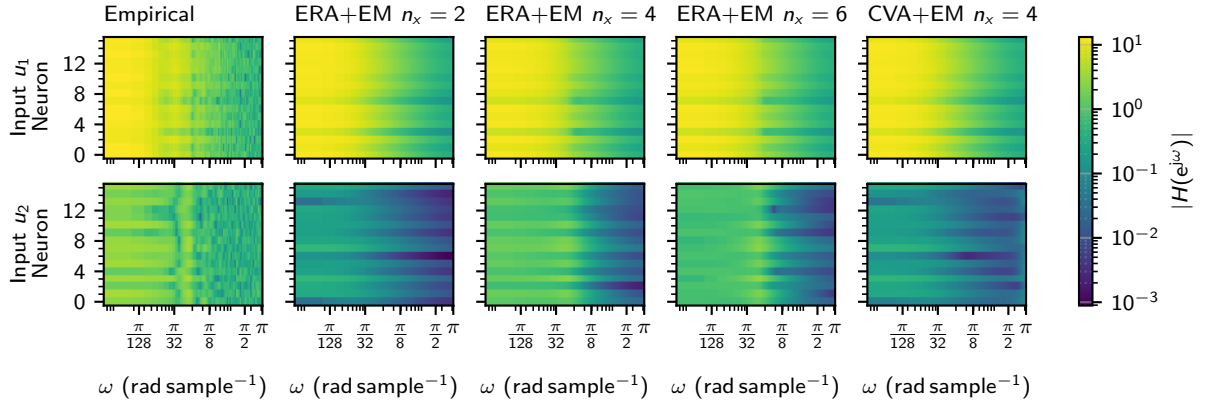


Figure 5: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $n_y = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample $^{-1}$ (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for $n_x \in \{2, 4, 6\}$ respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at $n_x = 4$, for estimator comparison at the chosen order. [Source](#)